BUREAU OF LAND MANAGEMENT
Financial Assistance (Cooperative Agreements)

ATTACHMENT A

PROJECT PROPOSAL

(Suggested Format)

Instructions: A Project Proposal must be submitted with the Standard Form (SF) 424 Application for Federal Assistance, for all BLM Assistance Agreements. Complete each section below. Use additional sheets as needed.

Person Submitting Proposal:	Elizabeth Leger, Shannon Swim, Tom Parchman
Date:	03/11/2024
Organization Name:	University of Nevada, Reno (UNR)
NOFO No.:	L24AS00081
Applicant Project Title (not the NOFO Title):	Maximizing diversity and minimizing selection during the seed cleaning and storage phase of native seed development
Estimated Period of Performance:	5 years
Proposed Project Location (include BLM District/Field Office and BLM POC as applicable):	Reno, NV
If project is to be awarded under a CESU please identify the appropriate CESU:	Great Basin CESU

DESIGNATED WATERSHED AREA:

This project will focus on optimal methods for processing seeds from across many watersheds, providing information that can be used to optimize seed cleaning and storage methods across multiple ecosystems. Our focus will be on native seed from the Great Basin and Mojave deserts, as there is extensive need for seed-based restoration in these regions, but general principles learned from our research will have nationwide applicability.

STATEMENT OF NEED:

Many factors contribute to the loss of habitat across the Great Basin and Mojave deserts, including the increased frequency, size, and severity of wildfires, altered precipitation cycles due to climate change, land conversion, and the spread of invasive species (Lovich & Bainbridge 1999, Davies et al. 2011, Svejcar et al. 2017, Smith et al. 2022). This has prompted the need to restore vast acreages within these arid and semi-arid ecosystems to mitigate vital habitat loss (e.g., Pilliod et al. 2017), especially for animals of particular concern like the Greater sage-grouse (*Centrocercus urophasianus*) and desert tortoise (*Gopherus agassizii*) (Pyke et al. 2015, Esque et al. 2021). Healthy native plant communities on public lands benefit many public interests, including agriculture, recreation, and biodiversity values, and maintaining these communities increasingly requires active management and seed-based restoration (Weiskopf et al. 2020). There are many steps in the process of obtaining genetically appropriate native seed, and seed limitations are often cited as reasons why managers cannot restore plant communities with desirable plants (NASEM 2023). A critical, yet perhaps underappreciated, step in ecological restoration is the cleaning and storage of wild-collected seed that either enter seed production projects or go directly to restoration sites. In support of the

Bureau of Land Management's (BLM) National Seed Strategy and Executive Order 14008, tackling the climate crisis at home, we propose to investigate the maintenance of diversity during the seed cleaning and storage steps that are essential aspects of restoring plant populations impacted by changes in fire, temperature, precipitation and other factors associated with climate change.

To-date, the US Forest Service (FS) Bend Seed Extraction (BSE) facility has processed and stored many of the Bureau of Land Management's (BLM) seed collections, as well as FS seed collections. As the need for more wildland seed collection increases, so does the demand for processing and storing these collections (NASEM 2023). More facilities are being created to serve these needs, and regional hubs have been recommended as a strategy to increase seed availability (NASEM 2023). Our region has one such hub, the Nevada Native Seed Partnership, and our group at the University of Nevada, Reno serves this partnership by providing research support in native plant restoration, along with native seed processing and storage capabilities. We focus specifically on processing and storing wild seed collections and smaller agricultural seed lots, including species that are difficult to handle and require extra attention. As a seed cleaning facility at a research institution, we are well-poised to not only provide clean seed to the restoration pipeline but to train students (graduate and undergraduate) and increase our knowledge of best practices while we provide seed services. With the increase in regional hubs, it is essential to understand how different processing methods might influence genetic and phenotypic variation in seed lots, and we are excited to use our facility to begin to answer these questions.

The need to understand the genetic and phenotypic consequences of various steps of the native seed supply chain has been long recognized (Hamilton 1994; Espeland et al. 2017). Genetic drift resulting from field collection and farm increase can reduce diversity and elevate inbreeding, and selection from cultivation could lead to evolutionary divergence in farm increased seed (Dyer et al. 2016, Hamilton et al. 2020). Recent empirical studies have documented phenotypic shifts occurring during the farm increase of native seed (Foxx and Kramer 2021, Pizza et al. 2021, Conrady et al. 2023), and others have evaluated the extent to which *ex situ* collection practices maintain genetic variation (Braasch et al. 2019; Backs et al. 2021, Zumwalde et al. 2022). However, to our knowledge, **none have investigated shifts in phenotypic or genetic variation that might occur through cleaning and few have investigated seed storage of native species, which is our focus here.** With increases in the affordability and speed of DNA sequencing, genome-level data for large numbers of individuals and populations stands to immensely improve our understanding of how genetic variation is shifted by seed processing (Breed et al. 2019). Additionally, our knowledge of the importance of particular seed and seedling traits is growing (Leger and Baughman 2015, Leger et al. 2019, Agneray et al. 2022), so we can understand not only if seed handling impacts diversity, but whether important traits such as seed size, emergence time, and early seedling traits (such as root allocation) are changing in non-optimal directions. There is information on long-term erosion of diversity during seed storage for crop species (e.g., Niedzielski & Puchalski 1998; Shelar et al. 2008), and while we know that seeds of different size are differentially susceptible to death during storage, we don't yet know how seed death impacts genetic diversity in native species, though modeling efforts indicate that they are susceptible to genetic loss (Richards et al, 2010). Minimizing loss or changes in diversity should be a priority for seed processing and storage, and an achievable goal, once we increase our knowledge of the susceptibility of particular collections to inadvertent selection and genetic drift during this stage.

Background and hypothesized results

Field collected seeds are typically cleaned to produce restoration-ready seed lots, and then stored under conditions designed to protect seed viability (Pedrini et al. 2020; Frischie et al. 2022).

However, because seed phenotypic variation is under genetic control and varies extensively in natural populations, the decisions made during cleaning could shift genetic variation from that represented in natural populations. This could result in reductions in genetic diversity or inadvertent selection for seeds that are less likely to be effective in restoration plantings (Basey et al. 2015; Kulpa and Leger 2013; Pedrini et al. 2019). For example, during the seed cleaning process, choices have to be made about how aggressively to pre-treat seeds, what machinery to use to extract and finish seeds and what degree of inert material is acceptable in batches of cleaned seed. This last step, dealing with inert material, we feel is in particular need of additional consideration, as the methods that lead to the purest clean seeds inevitably result in some seed loss.

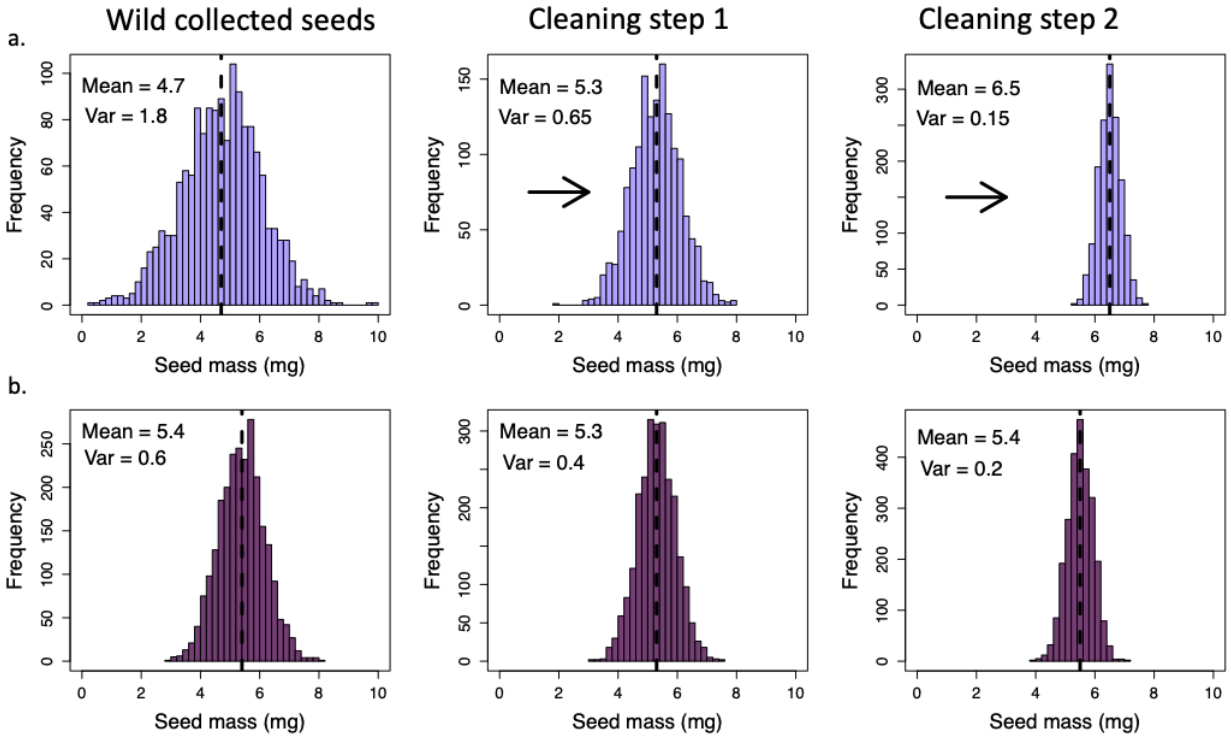


Figure 1. Example of potential changes in seed mass during the seed cleaning process. It is possible that loss of genetic variation and directional shifts in phenotype are highest in collections with naturally large variation in seed size (a), and lower in collections with lower variance in seed size (b).

We expect that wild-collected seeds will contain the greatest diversity in seed mass, and that for most species, variation in seed mass will decrease as processing removes inert material (Figure 1). If inadvertent selection is happening during the seed cleaning process, we expect that it would be due to the loss of smaller seeds (Figure 1a), though larger seeds could also be lost for species with seeds that are not easily extracted from inert material, as they could be lost during removal of larger inert material. We expect that the magnitude of seed mass change will differ by species and seed lot, with lots containing a relatively large range of seed sizes experiencing larger shifts (Figure 1a) than less variable lots (Figure 1b). If this hypothesis is correct, populations with high variation in seed mass (something that could be assessed during the early stages of seed cleaning) may benefit from cleaning to lower purity levels, accepting more inert material in cleaned seed to maintain diversity.

We expect that any changes in seed mass will have impacts on phenotypic measures, as seed size is known to affect germination timing and early seedling characteristics (e.g. Zhang et al. 2014), though we are not sure whether there will also be underlying genetic shifts. Changes in seed mass, such as

the loss of small seeds during cleaning, could either result in a) no underlying change in genotype if individual plants make a wide range of seed sizes (Figure 2a), or b) changes in genetic variation (Figure 2b), if some maternal plants make overall larger seeds while others make overall small seeds.

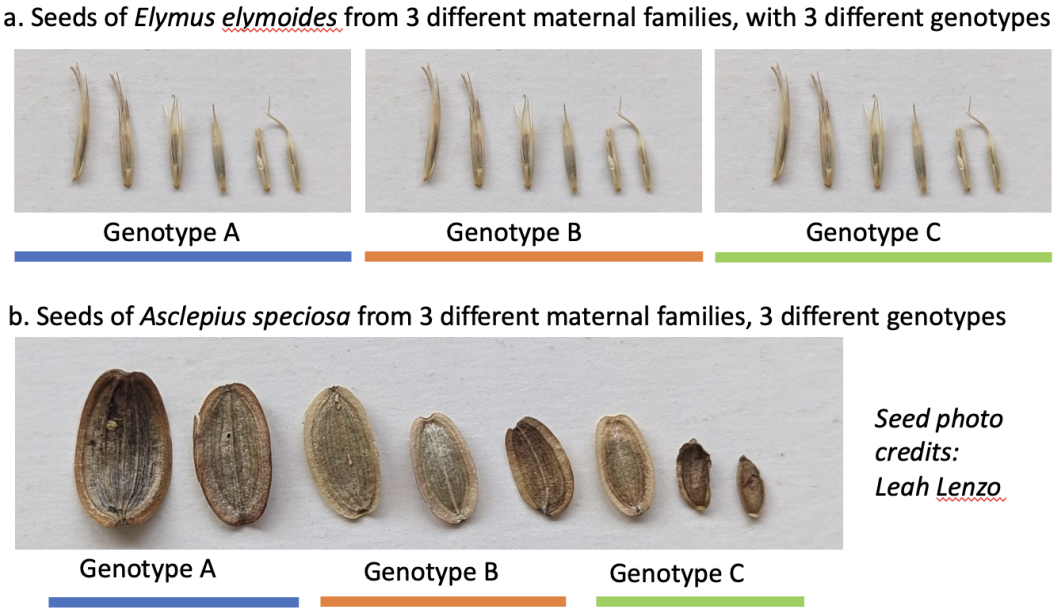


Figure 2. In (a) each hypothetical maternal family makes a range of seed sizes, so a loss of small seeds may affect phenotypic diversity, but not underlying genetic diversity, as loss is equal across maternal plants. In contrast, in (b), there is hypothetical genetically-based seed size variation among individual plants, so loss of small seeds during the cleaning process would shift both phenotypic and genetic variation. Genotype C may be adapted to particular growing conditions, but seed processing may reduce or eliminate these individuals from the population.

Because seeds are living things, storage is an important consideration for longevity and maintenance of diversity. Humidity and temperature are known to affect seed longevity (Harrington & Kozlowski 1972). While survival during storage is more studied than effects of seed cleaning, there are still few studies that quantify phenotypic or genetic shifts in seed during storage (Clorr et al., 1976; Richards et al., 2010; Mira et al., 2019; Genna et al. 2020), with most finding that smaller seeds tend to have shorter longevity during storage. We expect to see this same pattern in our seed storage trial, with differences among species in the rate of death and responsiveness to inert material. For example, *Artemisia tridentata* is known to have short-lived seeds that are particularly sensitive to the presence of inert material (Karrfalt & Shaw, 2013). However, what is unknown is whether we are losing particular genotypes as we lose seed viability in this species. This will depend on the degree to which seed loss is even across all maternal plants, or genotype-specific. In this proposal, we will set up seed samples in a long-term storage design that will be necessary to answer this question, with genetic sequencing and phenotyping to be accomplished in a future project, after a sufficient storage period.

TECHNICAL APPROACH:

Using a tiered approach (Figure 3), we will quantify seed processing and storage effects on a range of native plant species, and set up experiments required to understand the consequences of storage on the diversity and longevity of seeds.

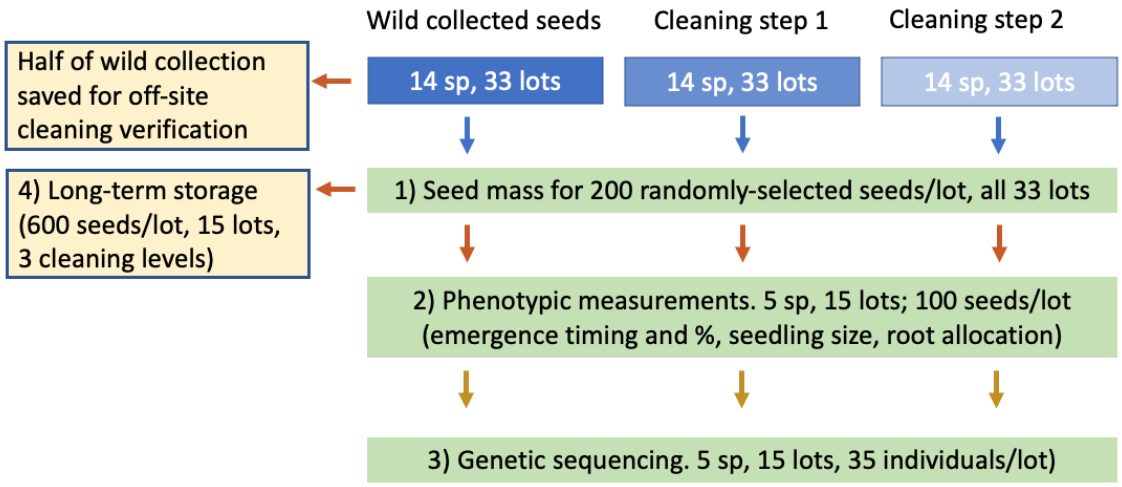


Figure 3. Experimental design for detecting 1) seed mass changes in 14 species, 2) phenotypic changes, 3) genetic shifts, and 4) phenotypic shifts during long term storage for 5 target species. Half of wild collections will be saved, uncleaned, and sent to a second seed cleaning facility for verification if meaningful changes in mean and variance are observed.

We will work with our collaborators at the Walker Basin Conservancy to collect seeds for these experiments, following SOS protocols and acquiring necessary collection permits. Overall, we will work with 14 separate species and 33 separate lots (collections from unique populations), with collection activities staggered over the first three years of this project. During the cleaning process, we will assess changes in: 1) seed mass, 2) seedling phenotypes (emergence and seedling characteristics, 3) DNA sequence variation, and 4) set up a long-term storage experiment to detect change during seed storage. For each of these 14 species, we will collect seeds from three (5 workhorse species) or two populations (9 additional species). All 33 lots will undergo measurements of initial seed mass and mass following cleaning to two different purity levels to ascertain how cleaning may influence seed mass. Our five workhorse species (15 collections) will also be used in experiments to measure seedling phenotypes and genetic sequences to ascertain if shifts have occurred, and we will save half of these wild collections so they can be sent to a second seed cleaning facility for comparison if needed. These 15 collections will also be entered into the storage component of this study.

Below we describe our rationale for species selection, our approach for establishing best practices for seed cleaning and storage, as well as features of our facility and resources that position us for success in our research objectives.

Species selection and collection methods

We will do our most intense sampling on five workhorse species, including shrubs, forbs, and a native grass (Table 1). We picked these species for their utility in restoration, their prevalence on the landscape, our previous landscape genomics work, and to include a range of seed characteristics such as seed size, awns, and pappus. For our additional nine species, we will select from the BLM's priority seed collection list, working with our collaborators at WBC to opportunistically collect from species that are having good seed production years. Seed for these cleaning experiments will be collected by WBC, and after our experiments are completed, any remaining seeds will be incorporated into restoration projects (through partners like the Nevada Department of Wildlife) or provided to growers for agricultural increase (through our Nevada Department of Agriculture partners). We have worked successfully with WBC in the past. They will obtain the appropriate

permits prior to collection, and will follow SOS seed collection protocols (e.g., accurate species identification, collecting from at least 50 individuals in a population, documenting number of individuals sampled, and post-collection handling etc.). Additionally, their collection crews attend the Great Basin SOS annual training.

Table 1. Five workhorse species that will be utilized in our phenotypic, genetic diversity and storage analyses, along with examples of additional candidates for use in the seed mass analysis.

Workhorse species	Potential additional species
Shrubs:	**Shrubs:**
1. *Artemisia tridentata*	*Atriplex sp.*
2. *Krascheninnikovia lanata*	*Ericameria nauseosa*
	Grasses:
Grass:	*Elymus elymoides*
3. *Achnatherum thurberianum*	*Aristida purpurea*
	Forbs:
Forbs:	*Cleome lutea*
4. *Balsamorhiza sagittata*	*Crepis occidentalis*
5. *Chaenactis douglasii*	*Sphaeralcea sp.*

We will collect seeds in a staggered design over three years, with three populations of two workhorse species and two populations of three additional species in year 1, three populations of three remaining workhorse species and two populations of three additional species in year 2, and the final two populations of three additional species in year 3.

1. Changes in seed mass during seed cleaning

For each of 33 lots from 14 species, we will measure the seed mass of 200 randomly-selected seeds from each of the three different levels of seed processing: uncleaned, wild-collected seed lots, seeds cleaned to between 30-60% purity (low-medium; set to stop at the point where cleaning methods are losing viable seeds), and seeds cleaned to at least 85% purity (high). Before processing, each seed lot will be assessed to determine the best extraction and finishing practices. Some factors that will influence our cleaning decisions include: species, collection size, and the amount and type of extraneous material. For example, for a fragile species like *Aristida purpurea,* we would hand extract using sieves. For species that are not prone to damage, we will extract seeds using our Westrup LA-H Brush Machine to separate seeds from associated plant material. During the extraction phase, seed loss is easily mitigated, as removal of material is minimal, but seed damage is the greatest risk. However, during the finishing phase, when seeds are separated from chaff, the risk of damage is minimal but the risk of non-random seed loss is high. We will thus develop our strategies based on the individual seed lot assessments. During cleaning we will follow our typical protocols, which involve sampling at each step and x-raying a sample to determine whether we are losing filled seeds.

Seeds will be individually weighed using our undergraduate work force (they report that they like seed weighing, which is surprising but great news for this project), and a subset of seeds will enter the phenotype and genetic analyses experiments. We are taking the approach of surveying many species at this stage, with less sampling of among-population variation, as we are very interested in understanding whether changes in seed mass during seed cleaning is a common phenomenon (Figure 1a) or less of a cause for concern (Figure 1b).

We will take a staggered approach to these measurements to spread out the workload. Specifically, in year 1, we propose to focus on weighing seeds from two workhorse species (6 lots) and three additional species (6 lots); in year 2, three workhorse species (9 lots) and three additional species (6 lots); and in year 3, two additional species (4 lots). If we observe differences of over 5% in average seed mass, or reductions in variance of greater than 30% in the highest purity lots, we will send uncleaned lots to a 3rd party seed cleaner for comparison, and returned seeds will be weighed to understand if change is consistent across facilities.

2. Changes in other phenotypes during seed cleaning

This experiment will use seeds that have been weighed during experiment 1. Specifically, 100 seeds from 3 lots for each of 5 workhorse species (15 total lots) cleaned to 3 different purity levels will be retained to use for greenhouse experiments, which will determine whether other important seed and seedling traits are shifting during the cleaning process.

To measure seed and seedling characteristics, we will plant individual seeds using our typical lab methods (e.g., Agneray et al. 2022), and carefully track emergence timing and a range of traits for seedlings grown for 10-days after emergence (total size, root allocation, root length, and other root morphological characteristics). We will again take a staggered approach to these grow outs, focusing on two workhorse species in the first growing season and three workhorse species in the second growing season. Briefly, we will plant seeds directly into a mix of Nevada topsoil and decomposed granite, using Ray Leach Super Cell pots (https://stuewe.com/) that fit into 98-pot trays. We will hold our greenhouse to fluctuate with ambient temperatures, allowing them to get cold (down to 2° C) in the evenings and warming (15-20° C) in the day. Plants will be randomized after planting, and watered daily during seedling emergence, with a light mist to keep the soils moist. We will carefully monitor emergence, either daily or, for rapidly-germinating species (such as *K. lanata*), on a 2x a day timeline. For species that require cold stratification before germinating (*B. sagittata, C. douglasii*), we don't plant seeds directly, but germinate them first in petri dishes in our growth chambers, holding moist seeds at 2° C for up to 12 weeks. These dishes are checked on a daily basis, with seeds buried in the soil when we observe radicles emerging.

We will harvest individual plants 10 days after seedling emergence from the soil. Seedling characteristics measured at this age have been shown to predict survival in competitive restoration situations for multiple species (Leger et al. 2019, Agneray et al. 2022). Seedlings will be gently washed from the soil (Figure 5), roots will be separated from shoots, and shoots will be scanned using the WinRhizo system (Regent Instrument Inc., Ville de Québec, QC Canada). Roots and shoots will then be dried

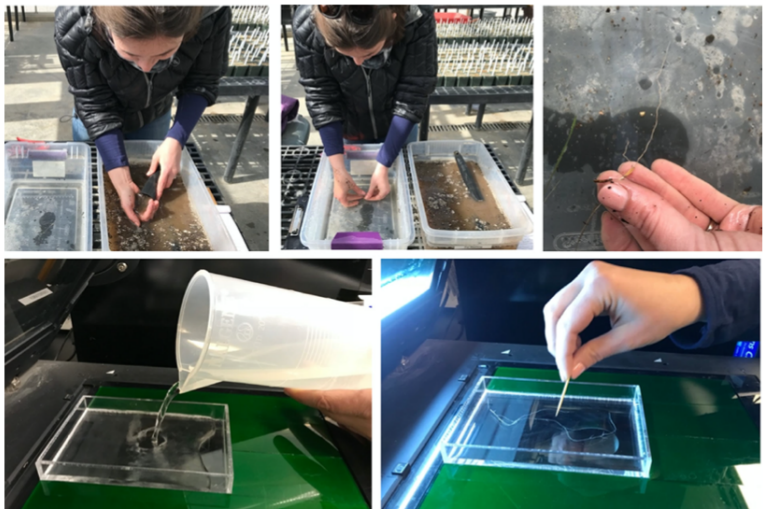


Figure 5. Reprinted from Leger et al. 2021. Top panels: gently removing seedlings from pots, rinsing to remove soil particles. Bottom panels: scanning involves floating cleaned root systems in water and scanning with WinRhizo software.

and weighed. These measurements will be used to calculate root to shoot ratio, specific root length (a measure of root thickness), total root length, the number of root tips (important areas for absorbing nutrients), and total size.

3. Shifts in genetic variation during seed cleaning

Using plants germinated in step 2, we will randomly-select dried material from 35 individuals from 3 lots each of the 5 workhorse species. Using DNA sequencing data generated as described below, we will compare genetic variation for each seed lot across three levels of processing: the wild-collected seeds and seeds after levels one and two of cleaning. Past studies have illustrated sample sizes of 15-25 individuals and ≥ 8,000 genomic loci are more than sufficient for estimating population level parameters specified below with high precision and accuracy (e.g., Willing et al. 2012; Nunziata et al. 2017; Nazarano et al. 2017). Our sampling of individuals and loci, to be executed as described below, will exceed these thresholds and ensure sound parameter estimation for understanding shifts in genetic variation.

DNA will be extracted using Qiagen DNeasy plant kits (Qiagen, Inc.), and screened for quantity and quality using a QIAxpert quantitative electrophoresis device. We will generate population genomic data using a reduced representation sequencing approach commonly executed with plant material in the Parchman lab (e.g., Uckele et al. 2021, Faske et al. 2021). This method reduces genomic complexity by digesting DNA with restriction enzymes and subsequently PCR amplifying and size selecting a subset of the fragment distribution for high-throughput sequencing on the Illumina platform. Full details on laboratory methods can be found in Parchman et al. (2012). Each library will contain 450-600 uniquely barcoded individuals and libraries will be sequenced with the highest output Illumina chemistry at the facility that is currently most cost effective (each lane of S2 NovaSeq: generates 1.9 billion short DNA reads). This approach offers a cost-effective method for generating population genomic data that can accurately quantify genome-wide levels of genetic diversity, as well as genetic differentiation among source populations and seed lots that have been cleaned to different purity levels.

We will use computational methods for sequence assembly and variant calling, together with a previously developed series of Perl, Python and Linux tools, to genotype Single Nucleotide Polymorphisms (SNPs). These analyses are routinely utilized in the Parchman lab and will be run on the lab's computing resources (see Facility and resources below). Based on recent use of these methods for the candidate species proposed herein, we expect genomic data spanning 20,000-40,000 per individual across each species. We will calculate nucleotide diversity (p), heterozygosity (H_o and H_e), and inbreeding coefficients (F_{IS}) to compare genetic diversity among source populations and seed cleaned with to different purity levels. As both drift and selection likely operate to some degree during seed cleaning, differentiation among seed cleaned at different levels will be assessed using standard metrics of population genetic differentiation (F_{ST}), and by Principal Component Analyses including cleaned seed along with previously generated genotypic data spanning the natural populations of these species (e.g., Faske et al. 2021; Osuna-Mascaróet al. 2023). Combined with experiment 2, this will result in a peer-reviewed paper that presents the first look at genotypic and phenotypic shifts during seed cleaning; if warranted, these publications may be separated.

4. Storage effects on native seed diversity

To initiate longer-term work aimed at understanding how storage duration might influence seed lot variation, we will store 600 seeds from each of the 15 lots from our five workhorse species (three lots each) at each level of purity. Seeds will be stored in one of our storage units set at the standard 5°C with humidity levels around 30% RH, which are considered the optimum conditions for shorter-term

storage of orthodox seed (Harrington & Kozlowski 1972; De Vitis et al., 2020). At each of three durations of storage (3, 5, and 10 years) we will ultimately assess changes in live seed mass and seedling characteristics for all 15 collections and 3 purity levels. Because we expect any genetic or phenotypic changes due to storage to be longer-term than shifts in seed cleaning methodologies, we will begin processing these sets in year 4 of this proposal, which will be the 3rd year of storage for collections from year 1. Due to the staggered nature of collections, we will subsequently sample seeds stored in year 2 in the 5th year of this project. Our goal is to sample them again at 5 and 10 years after their initial storage date. Our efforts here will produce a solid foundation for more in depth analyses of how storage duration influences variation, and set up opportunities to gain long-term knowledge, including whether seed death in storage leads to reductions in diversity or changes in genetic composition of stored seeds. Funding for these longer-term tests will be sought in future proposals, but it is essential that we store seeds now to allow us to understand long-term change.

Facility and resources

The UNR Leger Lab Native Seed Processing/Storage Center was created in 2017, funded by the Walker Basin Project, authorized and funded by Congress through Public Law 109-103, Section 208, with partial staffing support provided from BLM agreement L20AC00277 from 2020-2023. A current expansion of capabilities is being funded by Department of Interior Bipartisan Infrastructure Law Burned Area Response (DOI BIL BAR) funding. This facility increases Nevada's ability to process and store genetically appropriate native seed for habitat restoration projects. We have a variety of seed cleaning, conditioning, testing, and storage equipment (Table 1), and to-date, we have processed over 75 different Great Basin and Mojave species, stored over 1500 pounds of seed, and trained multiple technicians and students on processing and screening equipment. Our staffing includes a center manager and hourly undergraduate students, along with graduate students conducting research that relies on our facility and resources.

We are well positioned to archive and analyze the large amounts of genetic and phenotypic data to be generated for this project. Archiving and analyses of data will be executed on the Parchman lab cluster (3 compute nodes with 512 GB of RAM, and 110 TB of storage), and data will be independently backed up on UNR's Rosalind long-term data storage server (2 Petabyte Isilon storage system). We will additionally utilize UNRs high-throughput computing (HPC) cluster, which graduate students may use free of charge, to facilitate analyses of genomic data.

Table 1. UNR Leger Lab Native Seed Processing/Storage Center equipment

Current/ future[1] cleaning equipment	Testing equipment
Westrup LA-H Brush Machine	Creative Electron X-ray Machine
Missoula Dewinger	Achieva Germinator
Wintersteiger LD 180 Laboratory Thresher	Bolioptics Microscope
2 Mater Continuous Seed Blowers[1]	Rotronic moisture analyzer
Office Clipper Air Screen Separator	Current/ future[2] storage
Mater Small Vibratory Separator	72 ft^3 refrigerator for 5°C, 72 ft^3 freezer for -18°C
Various Sieves	1200 ft^2 walk in cold storage, 5°C[2]

[1]One additional continuous seed blower will be added to our facility from DOI BIL BAR funds.
[2]A 1200 cubic foot mobile walk-in unit will be added to our facility from DOI BIL BAR funds, adding 2,000 pounds to our 5°C cold storage capacity.

PROJECT MONITORING AND EVALUATION PLAN:

We will document in detail all aspects of the project, including cleaning protocols used, seed mass, phenologic, genetic, and storage data collection, with our results presented in at least two peer reviewed manuscripts and multiple presentations. Project PI Leger and co-PI Parchman will meet weekly with the PhD students working on this project to evaluate and assist with progress, data curation and analysis, and project write up. Co-PI Swim will oversee all seed cleaning activities and cleaning personnel, and assist with project design, training, and implementation.

Our timeline of activities, below, will serve as the basis of our annual reporting, and we will evaluate our success based on these milestones. If we need to adjust timeframes due to drought or other issues, we will include this in annual reports, though we hope that our methods are flexible enough to collect seeds for our purposes even in the driest years. If at any point we are unable to complete one of the listed tasks, we will consult with the BLM to identify an alternative approach and/or timeline.

TIMETABLE OR MILESTONES TO COMPLETE EXPECTED OUTCOMES

Milestone / Task / Activity	Start Date	Completion Date
Onboard Leger graduate student, conduct PhD work	9/2024	8/2029
Hire 3 undergraduate technicians/begin training (staggered in a 5 year position, 4 year position, and 3 year position)	9/2024	8/2029
Contract with WBC for staggered seed collection (33 total collections from 14 species)	9/2024	10/2027
Seed mass data collection for all 33 collections; subset of collections sent to 3rd party for cleaning verification if meaningful change is observed	12/2024	6/2028
Clean to two purity levels (low-med and high) for all 33 total collections	12/2024	12/2027
Onboard Parchman graduate student, conduct PhD work	1/2025	9/2027
Conduct after cleaning seed mass/ x-ray for each purity level data collection (33 total collections)	1/2025	6/2028
Enter seeds from workhorse collections and different purity levels into long-term storage experiment (15 total collections, 3 treatments, for a total of 45 lots stored)	12/2024	12/2027
Phenotypic data collection for workhorse collections	2/2025	6/2028
Genetic diversity experiment for workhorse species, DNA extraction, sequencing	2/2025	6/2028
Seed mass manuscript preparation	6/2026	12/2027
Findings reported through posters and presentations at NNSC (2 conferences)	2/2027	2/2029
Begin storage experiment Year 3 data collection (seed mass and phenology measurements)	12/2027	8/2029 and beyond
Phenotypic and genetic diversity peer-reviewed manuscript(s) preparation	6/2028	8/2029 and beyond
Annual report submission (5)	12/2024	12/2028
Final report submission (1)		12/2029

PUBLIC BENEFIT AND PROGRAM INTEREST OF THE BLM:

Getting the right seed, in the right place, at the right time is critical to successfully restoring our public lands, which support numerous public interests such as livestock production, recreation (hunting, wildlife viewing, etc.), and biodiversity. These lands also offer critical ecosystem services such as dust mitigation and water retention, and support wildlife and pollinator habitat, which the public relies upon to support our food system and conservation values. The BLM's SOS program takes great care to ensure that we are maintaining the needed diversity during collection to enhance the outcome of restoration activities. Our project aims to continue these efforts, asking whether seed cleaning and storage may erode diversity, and if so, identifying the steps that can be taken to minimize these shifts, highlighting species or other characteristics of importance. With the NASEM recommendations of regional restoration hubs, more and more independent seed cleaning operations like ours are being created and planned (we have already spoken to several folks who are working on this). Enhancing our understanding of the potential for phenotypic and genetic shifts during seed cleaning and storage is becoming of paramount importance to ensure that we are, in fact, providing the right seed. Additionally, our project will train undergraduate and graduate students, contributing to the future restoration workforce, including potential BLM employees, helping to ensure a knowledgeable workforce to maintain our public lands for the public's interest into the future.

QUALIFICATIONS/PAST PERFORMANCE:

Elizabeth Leger, PhD, is the PI of this project. Her background is in plant ecology and plant restoration, and she has been researching and implementing associated projects at the University of Nevada, Reno, since 2006. She has worked with the BLM on multiple agreements (several described below) which involved conducting a review and analysis of SOS collections across the US, describing the diversity and scope of collections along with species-specific analyses of how collections are distributed across potential habitat and restoration sites. She has also conducted work on the importance of seed characteristics in ecological restoration, including describing characteristics that increase seedling survival in restoration scenarios. PI Leger will be responsible for project progress, including annual reporting, and will supervise the PhD student working on seed and seedling traits. While there are no overload days included in the budget, her FTE commitment is approximately 15%. Contact information: lelizabeth@unr.edu, 775 784-7582

Since arriving at UNR in 2013, co-PI Tom Parchman, PhD, has been generating and analyzing population genomic data across a variety of species for research involving the evolutionary consequences of species interactions, the genetics of adaptation, and landscape genetic variation in western plant species. Through collaboration with PI Leger, this work has increasingly focused on genomic variation in Great Basin plants important to ecological restoration. The equipment, personnel, and computational expertise necessary for the completion of this project are established in the Parchman lab. Co-PI Parchman will lead the laboratory generation and analysis of high throughput sequencing data generated to assess genetic shifts occurring through various cleaning and storage methods, and will advise the PhD student that will be recruited to work on restoration genetics of Great Basin natives. While there are no overload days included in the budget, his FTE commitment is approximately 10%. Contact information: tparchman@unr.edu, 775 784-4484

Co-PI and Project Manager, Shannon Swim, MS, led the creation of the UNR Leger Lab Native Seed Processing/Storage Center in 2017 to help increase Nevada's native seed processing power. Since then, she has been managing all aspects of this center and has processed over 75 different species and stored over 1500 pounds of native seed for various organizations. Her dedication to this project will

be 25% FTE. For the first year and a half of this project the remaining portion of her salary (75%) will be covered by DIO BIL BAR funds, which fund our routine seed cleaning and management activities. We will be applying for additional DOI BIL BAR funds for 75% of her salary in the remaining years. She will be responsible for overseeing the cleaning, storage, and seed mass portions of this proposal. Additionally, she will conduct all the Leger lab graduate and undergraduate training. Contact information: swim@unr.edu, 775 784- 4111

Walker Basin Conservancy (WBC) is a local 501(c)(3) nonprofit organization whose goal is to preserve Walker Lake and also specialize in wildland seed collection. We have utilized their services for other projects. WBC sends their seed collection technicians to the Great Basin Annual SOS training, follows SOS collection protocols, and obtains appropriate seed collection permits. They are an invaluable contractor for this project as seed collection is an arduous task.

DNA Sequencing services: The Parchman lab has made use of numerous DNA sequencing of facilities. For this work, we will use the facility that is currently most cost effective, as expenses and platform outputs can change rapidly.

PI Leger and co-PI Parchman have cooperated on multiple assistance agreements with the BLM within the last three years. These include:

L16AC00318, Supporting the National Seed Strategy with research and technical support. Elizabeth Leger, PI. Thomas Parchman, Co-PI. $490,000
This multi-faceted project provided support for multiple projects related to native plant restoration and genetics. Objectives included: Improving access to plant distribution information in the Great Basin, collecting and increase seeds of native forbs not currently available in our region, describing patterns of genetic diversity, genetic structure, and gene flow for an important sagebrush steppe species, and associating these patterns with phenotypic variation and restoration success; testing the relative performance of restoration mixes composed of seed sources that differ in evolutionary history. This award resulted in 9 peer-reviewed publications, and partially supported 2 PhD students, one MS student, and two post-doctoral researchers, along with multiple undergraduates. All the graduate students and one post-doc went on to careers in the federal government.

L19AC00013, Using plant genetics to improve range restoration. Thomas Parchman, PI, Elizabeth Leger, Co-PI. $209,745
The objectives of this work were to describe patterns of landscape genetic variation across populations of multiple geographically and environmentally widespread plant species in order to: a) quantify population genetic diversity and connectivity, and b) to quantify the extent to which population genetic variation is associated with geography, environmental variation, and evolutionary history; use common garden studies to quantify phenotypic variation using seed collected across such populations in order to: a) describe phenotypic variation among populations and evidence for local adaptation, and b) quantify the degree to which population genomic variation is associated with phenotypic traits that are likely important to local adaptation and seedling establishment. This project resulted in 3 peer-reviewed publications, and partially supported 2 PhD students, one post-doctoral researcher, and multiple undergraduates.

L21AC10525, Opportunities to increase restoration in the Great Basin, Thomas Parchman, PI, Elizabeth Leger, Co-PI. $116,909 original award, $113,087 amendment 10/23.
This work aims to leverage high throughput sequencing data to evaluate how natural genetic variation may be modified through the process of *ex situ* field collection and farming practices used to increase seed, including asking if promising populations of native seeds can maintain their ability to establish from seed following farm increase. This work will involve collaboration with the BLM to

grow, sequence, and plant native species in Nevada's Great Basin ecosystems. With help of BFI Native Seeds, we have produced farm-increased restoration seed from three promising native sources of both *Elymus elymoides* and *Achnatherum thurberianum* as a basis for our analyses and as seed sources for future increase. We have collected seed from commercial cultivars and *ex situ* GRIN collections for five species, and are in the process of germinating seed and extracting DNA for high-throughput sequencing. We have conducted landscape genomic analyses to quantify evolutionary history and spatial genetic structuring across the Great Basin distributions for multiple species for which we aim to evaluate genetic variation in restoration seed, including *A. thurberianum* (Osuna-Mascaró et al. 2023), *Balsamorhiza sagittata* (Osuna-Mascaró et al. manuscript in prep), *Chaenactis douglasii,* and *Artemisia tridentata* (Faske et al. manuscript in prep).

L20AC00317, The role of seed size in improving success of Great Basin habitat restoration and vegetation management, Elizabeth Leger, PI. $132,000

This work aims to understand how seed mass and fill vary across both space and time to increase success in wildland seed collection, restoration ecology, and conservation biology, using datasets from the Bureau of Land Management (BLM)'s Seeds of Success (SOS) program, along with information from the US Forest Service (USFS)'s Bend Seed Extractory. We organized a dataset that includes 13,314 accessions of 2,668 unique taxa, along with gathering explanatory information such as phylogeny, ecoregion, soil type, climate, weather, and collection time, to ask how these predictors influence seed mass, as well as variation in seed size. PhD student Leah Lenzo (Prescott) has presented these results at seven separate venues, and is working on the first manuscript for submission.

L20AC00277, Plant and seed collection, cleaning, and curation in support of Great Basin habitat restoration and vegetation management, Elizabeth Leger, PI. $66,000

The goals of this project were to document the locations of native species in the Great Basin with herbarium specimens, and increase access to seeds needed for restoration-scale projects by cleaning and storing native seeds for a variety of users, over a 3-year period. Our herbarium curator added 1400, 4996, and 2770 each year, including voucher specimens for BLM's SOS program, and our curator assisted SOS teams with plant identification. All of these records are publicly available at https://intermountainbiota.org/, which can help anyone locate populations of particular native species This project also helped support our seed bank manager as we began processing seeds, and over the three years, we handled over 300 lots of over 30 species. This support was instrumental in keeping our curation and cleaning work going.

LEVERAGING OF RESOURCES

Our project will leverage infrastructure expansion, which is being funded by Department of Interior Bipartisan Infrastructure Law Burned Area Response (DOI BIL BAR; $426,363 in year 1 for equipment purchases and personnel), though there are no science activities supported through these funds, as we propose here. Our project will also leverage our existing seed processing equipment, including a Westrup L-H Brush Machine, Missoula Dewinger, Wintersteiger LD 180 Thresher, Office Clipper, Mater Continuous Seed Blower, Creative Electron X-ray machine, Rotronic Moisture Analyzer, Germinator, various sieves, Microscope, operational supplies such as bags, and our cold storage units, which was purchased with funds from theWalker Basin Project, which was authorized and funded by Congress through Public Law 109-103, Section 208.

OVERLAP OR DUPLICATION OF EFFORT STATEMENT

There are no overlaps or duplications between this application and any of our other Federal applications or funded projects, including in regard to activities, costs, or time commitment of key

personnel. We understand that if at any time we receive funding from another source that is duplicative of the funding we are requesting from the Bureau of Land Management in this application, we will immediately notify the Bureau of Land Management point of contact identified in this NOFO in writing.

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